$$A^2 - \text{tr}(A)A + \det(A)I = 0$$

$$A^2 - (0)A + -2I = 0$$

$$\underline{A^2 = 2I}$$

Char. eq^n : $\lambda^2 - 2 \rightarrow \lambda = \pm\sqrt{2}$

$$A^2 = 2I$$

$$A^{12} = 2^6 I \rightarrow A^{13} = 2^6 A$$

Eigen values of $A^{13} = 2^6 \times$ eigen values of A

$= 64\sqrt{2}, -64\sqrt{2}$